

submodules

Owned-submodule capabilities

Inventory of the discrete features each owned (own-org vasic-digital / HelixDevelopment) submodule under submodules/* provides to HelixCode. Assessed from each submodule’s README.md, exported package surface, and *_test.go presence (CONST-035 / §11.4.107 — evidence-based, no bluffs).

Wiring model (load-bearing). HelixCode’s inner Go module (helix_code/go.mod) replaces + imports only 3 submodules directly — helix_agent (dev.helix.agent), dag_orchestrator (dev.helix.dag), pipeline_runtime (dev.helix.pipeline). HelixCode source additionally imports a handful by module path (containers, helixspecifier, helixqa, concurrency, helixmemory, debate, lazy, llmsverifier, memory, llmprovider). A large set is wired **transitively** because helix_agent/go.mod requires ~35 own-org submodules — these are reachable through the agent dependency but NOT imported by HelixCode directly (Wired=partial). The remainder are present in the repo as equal-codebase submodules (CONST-051) but are NOT in HelixCode’s Go build graph at all (Wired=no). 📺 Video=no for every row (conductor owns video confirmation; nothing here is confirmed). Origin=native throughout (all own-org).

Area	Component	Feature	Dev	Wired	Real-use	Te
submodule	helix_agent	Ensemble multi-LLM agent service (response fusion)	done	yes	yes	un
submodule	helix_agent	ReAct/tool-calling agent runtime (consumed by HelixCode internal/agent)	done	yes	yes	un
submodule	helix_agent	Skill registry + debate + verifier wiring (umbrella, 1583 test files)	done	yes	unknown	un
submodule	dag_orchestrator	Agent-free DAG scheduler (topo dispatch, worker pool, retry/backoff)	done	yes	yes	un
submodule	pipeline_runtime	Staged streaming dataflow runtime (push operators + FBP backpressure)	done	yes	yes	un
submodule	helix_specifier		done	partial	unknown	un

Area	Component	Feature	Dev	Wired	Real-use	Te
		Spec-driven dev fusion (SpecKit 7-phase + TDD + GSD milestones)				
submodule	helix_specifier	10 power features (parallel exec, constitution-as-code, debate phases)	partial	partial	unknown	un
submodule	helix_memory	Unified cognitive memory (Mem0+Cognee+Letta+Graphiti fusion)	done	yes	unknown	un
submodule	helix_memory	Parallel cross-backend search + 3-stage fusion/dedup/re-rank	done	yes	unknown	un
submodule	debate_orchestrator	Multi-agent debate orchestration (consensus + dissent, LessonBank)	done	yes	unknown	un
submodule	debate_orchestrator	Aux packages (validation/audit/evaluation/reflexion/tools)	stub	partial	no	un
submodule	concurrency	Worker pools, priority queues, rate limiters, circuit breakers, semaphores	done	yes	yes	un
submodule	lazy	Type-safe lazy-init primitives (sync.Once generics)	done	yes	yes	un
submodule	memory	Mem0-style scoped memory + entity extraction + knowledge graph + leak detect	done	partial	unknown	un
submodule	llm_provider	LLMProvider interface + circuit breaker + health monitor + retry + lazy	done	partial	unknown	un
submodule	llms_verifier	LLM verification platform (existence/latency/streaming/vision/embeddings)	done	yes	unknown	un
submodule	llms_verifier	Single-source-of-truth model/provider metadata (CONST-036/037)	done	yes	unknown	un
submodule	containers	Container lifecycle (boot/compose/health) for infra-on-demand (§11.4.76)	done	yes	unknown	un
submodule	helix_qa	Autonomous QA / Challenge orchestration (test-bank runner) [QA infra]	done	yes	unknown	un
submodule	panoptic	Recording-validator / observation harness [QA infra]	done	partial	unknown	un
submodule	agentic	Graph-based agentic workflow engine (branch/checkpoint/self-correct)	done	partial	unknown	un
submodule	auth		done	partial	unknown	un

Area	Component	Feature	Dev	Wired	Real-use	Test
		JWT / API-key / OAuth2 / HTTP auth middleware / token store				
submodule	cache	Multi-backend cache (mem/Redis/PG) + distributed patterns + TTL/eviction	done	partial	unknown	unknown
submodule	storage	Object storage (S3/MinIO/local) + cloud credential mgmt	done	partial	unknown	unknown
submodule	vector_db	Unified vector store (Qdrant/Pinecone/Milvus/pgvector)	done	partial	unknown	unknown
submodule	embeddings	Text embeddings across 7 providers (OpenAI/Cohere/Voyage/Jina/Google/Bedrock)	done	partial	unknown	unknown
submodule	event_bus	Typed pub/sub bus + glob/prefix/metadata filtering + middleware	done	partial	unknown	unknown
submodule	messaging	Message-broker abstraction (in-mem/Kafka/RabbitMQ) + producer/consumer patterns	done	partial	unknown	unknown
submodule	streaming	SSE / WebSocket(rooms) / gRPC streaming / webhook(HMAC) / HTTP+breaker / Gin	done	partial	unknown	unknown
submodule	rag	RAG pipeline (3 chunkers, BM25+semantic+hybrid retrieve, MMR rerank, RRF fusion)	done	partial	unknown	unknown
submodule	tool_schema	Tool schema/validation/exec + 14 built-in safe tool handlers	done	partial	unknown	unknown
submodule	plugins	Plugin lifecycle (dep-ordered) + .so/process loaders + sandbox + output parse	done	partial	unknown	unknown
submodule	planning	AI planning algorithms (HiPlan + MCTS + Tree-of-Thoughts)	done	partial	unknown	unknown
submodule	optimization	Semantic cache + prompt compression + structured-output + SGLang/LlamaIndex/LangChain	done	partial	unknown	unknown
submodule	self_improve	RLAIF pipeline (reward models, feedback, policy/prompt optimization)	done	partial	unknown	unknown
submodule	benchmark	LLM benchmarking (SWE-bench/HumanEval/MMLU/GSM8K) + leaderboard	done	partial	unknown	unknown
submodule	llm_ops	LLMOps (continuous eval, A/B experiments, prompt versioning, alerting)	done	partial	unknown	unknown

Area	Component	Feature	Dev	Wired	Real-use	Te
submodule	mcp_module	MCP server+client (JSON-RPC 2.0 stdio+HTTP/SSE) + adapter registry	done	partial	unknown	un
submodule	models	Shared AI/LLM data types (LLMRequest/Response, MCP/ LSP/ACP, code-intelligence)	done	partial	unknown	un
submodule	formatters	Pluggable code-formatter registry + engine + cache + native-binary shims	done	partial	unknown	un
submodule	vision_engine	CV + LLM-vision UI analysis + navigation-graph construction	partial	partial	unknown	un
submodule	doc_processor	Doc-feature-map extraction + verification-coverage tracking (QA-oriented)	done	partial	unknown	un
submodule	security	Security tooling module (round-300 deep-doc) — needs deeper inventory	done	partial	unknown	un
submodule	red_team	YAML-driven adversarial-prompt fixture harness (defensive guardrail regression)	done	partial	unknown	un
submodule	llm_orchestrator	Headless CLI-agent orchestration (OpenCode/ Claude/Gemini/Junie/Qwen)	done	no	no	un
submodule	helix_llm	Distributed LLM service (OpenAI/Anthropic APIs, llama.cpp, RAG, ReAct, HTTP/ 3)	done	no	no	un
submodule	conversation	Infinite-context (event-sourcing) + LLM compression + LRU cache	done	no	no	un
submodule	database	Driver-agnostic DB (PG/ SQLite) + pool + migrations + repo + query builder	done	no	no	un
submodule	document	Document model (18-format detect, change tracking, JSON serialize)	done	no	no	un
submodule	filesystem	Multi-protocol FS client (SMB/ FTP/NFS/WebDAV/Local)	done	no	no	un
submodule	config	Config mgmt (JSON files, env binding, validation, 8 storage-protocol types)	done	no	no	un
submodule	i18n	i18n library (bundles, loader, HTTP middleware)	done	no	no	un
submodule	observability	Distributed tracing + metrics + structured logging + health + analytics	done	no	no	un

Area	Component	Feature	Dev	Wired	Real-use	Te
submodule	middleware	Reusable net/http middleware (requestid/logging/recovery/cors/chain)	done	no	no	un
submodule	rate_limiter	Rate limiting (in-mem + Redis) + HTTP middleware	done	no	no	un
submodule	recovery	Named circuit breakers + periodic health checks + resilience facade	done	no	no	un
submodule	watcher	Filesystem change monitoring (debounce, filters, handler chains)	done	no	no	un
submodule	background_tasks	Persistent task queue (PG) + worker pool + stuck-detect + DLQ + progress	done	no	no	un
submodule	skill_registry	Skill mgmt (load YAML/JSON/MD, register, execute, validate, store)	done	no	no	un
submodule	embeddings	(see embeddings row — module also standalone)	—	—	—	—
submodule	auto_temp	LLM temperature auto-tuning (multi-temp run + multi-judge scoring)	done	no	no	un
submodule	hyper_tune	LLM hyperparameter optimization (random/grid/Bayesian-lite)	done	no	no	un
submodule	i_llm	Structured-reasoning patterns (CoT/ToT/ReAct/few-shot/prompt-chain)	done	no	no	un
submodule	veritas	AI-truthfulness verification, fact-check, hallucination detection	done	no	no	un
submodule	leak_hub	System-prompt-leak detection + searchable archive	done	no	no	un
submodule	claritas	Defensive guardrail (leaked-prompt archive + extraction-attempt detector)	done	no	no	un
submodule	gandalf_solutions	Read-only Gandalf prompt-hacking solutions archive (research)	done	no	no	un
submodule	ouroborous	Self-referential AI-safety (recursive self-improve, runaway-loop detect)	done	no	no	un
submodule	normalize	Adversarial-input canonicalization (base64/leet/homoglyph/NFKC/ROT13...)	done	no	no	un
submodule	plinius_common		done	no	no	un

Area	Component	Feature	Dev	Wired	Real-use	Te
		Plinius shared lib (config validators, error types, gRPC client, i18n, types)				
submodule	toon	Token-Oriented Object Notation encode/decode	stub	no	no	un

55 features across 50 submodules.

Coverage-depth honesty

- **Shallow-inventoried (principal features only; deeper inventory warranted):** helix_agent (1583 test files — only top umbrella capabilities captured; internal package-level features not enumerated), helix_qa (646 test files, QA infra — single capability row), helix_specifier (37 tests, 10 power features collapsed to 2 rows), security (deep-doc module — only a single umbrella row; per-package security capabilities NOT enumerated, flagged as needs-deeper-inventory), panoptic (QA infra — single row).
- **Real-use is unknown** for nearly every transitively-wired (partial) and standalone (no) submodule because reachability through helix_agent's go.mod ≠ proof an end user exercises the capability via HelixCode; only the 3 directly-replaced modules + concurrency/lazy show yes.
- **Dev reflects README claims + test presence**, NOT runtime verification. toon is honestly stub (PENDING_IMPLEMENTATION per its own README); debate_orchestrator aux packages are stub (NotYetImplemented per ACK-STUB).
- **Tests column** marks unit baseline (all 70 own-org submodules have *_test.go); integ/e2e added only where README/dir evidence shows real infra/integration suites (cache, storage, vector_db, llms_verifier, helix_qa, helix_agent, containers, database, rate_limiter, helix_llm, observability-N/A).
- **Infra/tooling submodules excluded from feature rows per task scope:** docs_chain, challenges (and containers/helix_qa/panoptic included only as the capabilities HelixCode consumes, marked QA/infra).

Sources verified 2026-06-22: submodules/* , helix_code/go.mod , submodules/helix_agent/go.mod , each submodule's README.md + *_test.go

REPO-STATE-DERIVED (per §11.4.99 the sources are the cross-referenced repo trees, following the docs/ARCHITECTURE.md precedent — no external service documented). Cross-referenced against the live tree on 2026-06-22: - **submodules/* = 67 directories** present. The doc states “all 70 own-org submodules have *_test.go” / “70+”; the live directory count is **67** — a minor count drift to reconcile on the next revision (the per-row enumeration itself reflects the actually-present submodules). - **Wiring model confirmed at the build-graph level:** the doc's load-bearing claim that HelixCode's inner module directly replaces+imports only 3 submodules (helix_agent/dev.helix.agent, dag_orchestrator/dev.helix.dag, pipeline_runtime/dev.helix.pipeline) while ~35 are reachable transitively via helix_agent/go.mod is verifiable against helix_code/

go.mod + submodules/helix_agent/go.mod (the cited Wired=partial rows). -
No external-service version claim in this doc → no §11.4.99 staleness check applies;
the evidence is each submodule's README + exported surface + tests.